

LAB 2

# VLAN Configuration & Network Segmentation

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# LAB OVERVIEW

## TASKS

1

Configure VLAN 10 (Students) and VLAN 20 (Faculty) on switches

2

Assign switch ports to the respective VLANs

3

Set up subinterfaces on the router to enable inter-VLAN routing

4

Verify VLAN communication by testing connectivity within VLANs and between VLANs

## HARDWARE

1X

Cisco ISR4331 Router

1X

Cisco 2960-24TT Switch

4X

PC End Devices

5X

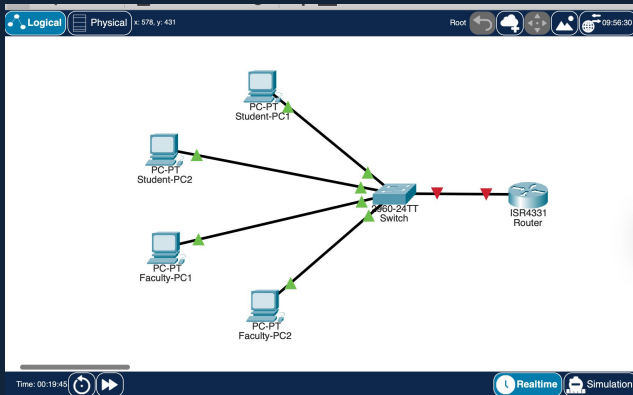
Straight-Through Cables

## SOFTWARE

CISCO Packet Tracer

# VLAN ASSIGNMENT

Text



VLAN Port & IP Assignment Table

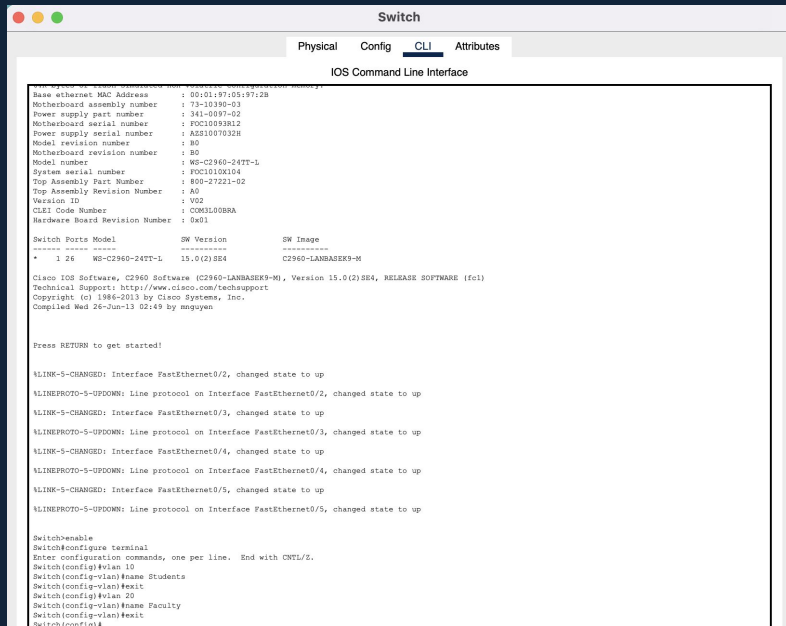
| DEVICE      | VLAN    | IP ADDRESS    | PORT  |
|-------------|---------|---------------|-------|
| Student-PC1 | VLAN 10 | 192.168.10.10 | Fa0/2 |
| Student-PC2 | VLAN 10 | 192.168.10.11 | Fa0/3 |
| Faculty-PC1 | VLAN 20 | 192.168.20.10 | Fa0/4 |
| Faculty-PC2 | VLAN 20 | 192.168.20.11 | Fa0/5 |

Subnet Mask: 255.255.255.0 · VLAN 10 GW: 192.168.10.1 · VLAN 20 GW: 192.168.20.1

Each device was assigned a unique static IP within its VLAN subnet, with the default gateway pointing to the corresponding router subinterface: 192.168.10.1 for Students and 192.168.20.1 for Faculty

# SWITCH VLAN CONFIGURATION

## Switch — CLI Interface



```
Switch
-----
Physical  Config  CLI  Attributes
-----
IOS Command Line Interface

Base ethernet MAC Address      : 001019710519712B
Motherboard assembly number    : 73-10390-03
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC10093812
Power supply serial number     : A021007032H
Model revision number          : 80
Motherboard revision number    : 80
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010K104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A5
Version ID                     : V02
CLEI Code Number               : COM3500RA
Hardware Board Revision Number : 0a01

Switch Ports Model        SW Version        SW Image
-----
*  3 24  WS-C2960-24TT-L  15.0(2)SE4       C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:43 by ngyuyen

Press RETURN to get started!

VLINE-S-CHANGED: Interface FastEthernet0/2, changed state to up
VLINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
VLINE-S-CHANGED: Interface FastEthernet0/3, changed state to up
VLINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
VLINE-S-CHANGED: Interface FastEthernet0/4, changed state to up
VLINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
VLINE-S-CHANGED: Interface FastEthernet0/5, changed state to up
VLINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Students
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Faculty
Switch(config-vlan)#exit
Switch(config)#
```

## Create VLANs 10 & 20

Entered global config mode using configure terminal command and created VLAN 10 named "Students" and VLAN 20 named "Faculty" using the vlan and name commands.

# PORT-TO-VLAN ASSIGNMENTS

## Ports Assigned to VLAN 10 (Students)

```
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
```

For VLAN 10 (Students): interfaces Fa0/2 and Fa0/3 were configured with switchport mode access followed by switchport access vlan 10, placing Student-PC1 and Student-PC2 exclusively in the Students segment.

## Ports Assigned to VLAN 20 (Faculty)

```
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
```

For VLAN 20 (Faculty): interfaces Fa0/4 and Fa0/5 were configured with switchport mode access followed by switchport access vlan 20, placing Faculty-PC1 and Faculty-PC2 exclusively in the Faculty segment.

# TRUNK CONFIG & VLAN VERIFICATION

## Trunk Port (Fa0/1 → Router)

```
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

## show vlan brief Output

```
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Students               active    Fa0/2, Fa0/3
20   Faculty                active    Fa0/4, Fa0/5
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

## Trunk Port - Router

FastEthernet 0/1 was set to switchport mode trunk telling the switch to tag outgoing frames with 802.1Q VLAN IDs so it can identify which subinterface to forward each frame through.

## VLAN 10 — Students Confirmed

The show vlan brief output confirms VLAN 10 "Students" is active with Fa0/2 and Fa0/3 assigned. This verifies Student-PC1 and Student-PC2 are correctly isolated within the Students segment at the switch level.

## VLAN 20 — Faculty Confirmed

VLAN 20 "Faculty" is confirmed active with Fa0/4 and Fa0/5 assigned. Faculty-PC1 and Faculty-PC2 are correctly placed in the Faculty segment.

# ROUTER SUBINTERFACE CONFIG.

## Subinterface for VLAN 10

Created subinterface G0/0/0.10, set encapsulation dot1Q 10 so the router recognizes VLAN 10-tagged frames, then assigned IP 192.168.10.1/255.255.255.0.

## Subinterface for VLAN 20

Created subinterface G0/0/0.20 with encapsulation dot1Q 20 and IP 192.168.20.1/255.255.255.0. This gateway IP enables Faculty PCs to reach the router and communicate with devices in the Students VLAN via inter-VLAN routing.

## show ip route Command

Output showed both VLAN networks as directly connected through the router subinterfaces.

## Router-on-a-stick Set Up

### VLAN 10 Subinterface

```
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
```

### VLAN 20 Subinterface

```
Router(config)#interface gigabitEthernet 0/0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
```

### show ip route output

```
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20
```

# PC STATIC IP CONFIGURATION

Each PC was manually configured with a static IPv4 address within its respective VLAN subnet. Student PCs use the 192.168.10.0 network and Faculty PCs use 192.168.20.0. The default gateway was set to the corresponding router subinterface IP to ensure proper inter-VLAN communication.

| DEVICE      | IP ADDRESS    | GATEWAY      | VLAN    |
|-------------|---------------|--------------|---------|
| Student-PC1 | 192.168.10.10 | 192.168.10.1 | VLAN 10 |
| Student-PC2 | 192.168.10.11 | 192.168.10.1 | VLAN 10 |
| Faculty-PC1 | 192.168.20.10 | 192.168.20.1 | VLAN 20 |
| Faculty-PC2 | 192.168.20.11 | 192.168.20.1 | VLAN 20 |

Subnet Mask: 255.255.255.0 · 192.168.10.0/24 (Students) · 192.168.20.0/24 (Faculty)

## Student-PC1 — IP Configuration Panel

The screenshot displays the 'Student-PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing the 'FastEthernet0' interface. The 'Static' radio button is selected for IP configuration. The settings are as follows:

|                            |                                         |
|----------------------------|-----------------------------------------|
| Interface                  | FastEthernet0                           |
| IP Configuration           |                                         |
| <input type="radio"/> DHCP | <input checked="" type="radio"/> Static |
| IPv4 Address               | 192.168.10.10                           |
| Subnet Mask                | 255.255.255.0                           |
| Default Gateway            | 192.168.10.1                            |
| DNS Server                 | 0.0.0.0                                 |
| IPv6 Configuration         |                                         |



# CONNECTIVITY TESTING

## Ping Within VLAN 10 (Students)

```
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

## Ping Within VLAN 20 (Faculty)

```
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4/4

Packets Received

0%

Packet Loss

3/3

Hosts Reached

## Ping Between VLANs (Inter-VLAN Routing)

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

# CHALLENGES FACED & SOLUTIONS

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## CHALLENGE

Forgot to set port Fa0/1 to trunk mode, so only one VLAN's traffic was passing through.

## SOLUTION

Used `show interfaces trunk` to verify, then ran `switchport mode trunk` to fix it.

## CHALLENGE

Forgot to add the encapsulation dot1Q command on the router subinterfaces, so the router wasn't recognizing VLAN traffic.

## SOLUTION

Matched each subinterface's dot1Q encapsulation to the correct VLAN ID on the switch.

## CHALLENGE

The first ping between VLANs showed packet loss and timed out.

## SOLUTION

Learned this is normal – It was just ARP resolution. The following pings all came back with 0% packet loss.

# LESSONS LEARNED

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## 01

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### VLANs Separate Networks

VLANs typically allow a single physical switch to be spread into multiple logical networks.

This helps keep different groups like students and faculty in separation while still using the same hardware.

## 02

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### Trunk Ports Carry Multiple VLANs

A trunk port allows traffic from multiple VLANs to travel over one connection between devices like a switch and a router. This is important for sending tagged VLAN traffic to the router for routing.

## 03

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### Router-on-a-Stick Enables Inter-VLAN Communication

Router subinterfaces with dot1Q encapsulation allow devices in different VLANs to communicate. The separate VLAN uses its own subinterface and gateway IP address so the router can route traffic between them.

# REAL-WORLD APPLICATIONS

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## Separating Different Users Within an Organization

Organizations such as universities, offices, and hospitals may need to separate different groups of users while using the same physical network.

## Hospitals

Isolate sensitive patient information, medical records, and preventing unauthorized access to comply with HIPAA

## Universities

Separating network traffic between students, faculty, and administrative staff to improve network security, avoid data breaches, and protect sensitive personal information